

Masters in Biostatistics

Portfolio of Evidence of Learning –Biostatistical Consulting and Collaboration 2019

Why are we asking you to create a portfolio of evidence?

This module is primarily applied, with the aim of equipping you with the skills required to effectively collaborate and consult with researchers. It is difficult to summatively assess the skills you have learnt in the module using traditional examination format. The move towards learning through assessment, and taking responsibility for your own learning lends this module to a more relevant way of assessment. We feel that a portfolio of evidence will help you to better use the knowledge gained through lectures, readings, and during the practical parts of your internship by reflecting on your learning process. It also provides us a more valid way of assessing what you have learnt.

Purpose:

- To demonstrate individual growth and learning throughout the module and internship
- To document your reflections on the learning processes in this module and internship
- To create a product for holistic assessment of this module

Format:

The format of this portfolio will be similar to the Student Report for your internship report, but will concentrate solely on the consulting and collaboration part of your internship (practical) as well as the lectures in the first semester.

The aim of this report is to give the reader your perception of what you learned, and how you learned it. In the report, you should discuss your perceived change in knowledge and skills as a result of undertaking the internship, and your contribution to the organization. In general, students should also highlight areas such as difficulties in determining appropriate analysis methods, useful sources of assistance other than the supervisor, and statistical consulting communication issues. The report should follow the format below:

Table 1 Format of the portfolio of evidence of learning^a

Introduction/Background • About 1-2 page narrative • Give general overview of your placement organisation (<i>do not include sensitive information</i>) • Explain your role and responsibilities regarding collaboration and consultation during the internship
Learning Objectives • Insert matrix of learning outcomes and tasks^b • Describe how you have met your learning outcomes
Relation to coursework ^c • About 1-2 page narrative or table • Discuss how your consulting work related to specific courses in the MSc Biostatistics programme
Work outputs • As many pages as needed; narrative, table, or some other organised forms • List and describe the consulting/collaborating you were involved in for the placement organisation

• Describe consulting/collaboration outputs, for example: input into protocols/grant applications, cleaned dataset, analysis reports, journal manuscripts, presentation slides, etc.. • Include case reports of interesting consultations.
Reflection • About 2-3 pages narrative - o What did you learn? - o Why was this learning significant? - o What would you have done differently? - o Identify any challenges you experienced and how you handled them.

^a Report should be 10-15 pages long, Arial font 11 in text and 10 in tables.

^b Select learning outcome of the module and identify specific activities through which you satisfied the learning outcomes.

^c Include 3 columns: (1) Indicate activity/task, (2) Identify two most relevant modules and (3) brief explanation of relevancy.

Content guidelines (artefacts):

- Case reports of interesting consultations during the placement
- Seminar presentation (video, slides etc)
- Reflections * on how coursework, assignments, discussions with convenors and supervisors, or feedback impacted on and helped shape the learning experience.
- Any other evidence of learning

Format:

- Cover page
- Table of contents
- Pages numbered
- Hard copy or electronic, or combination, or website – whichever format best suits your artefacts best
- Be creative and use different media for different artefacts

Assessment:

Assignment 1 (10%)

Assignment 2 (10%)

Rubric 3**Overall impression of Portfolio**

Criteria	Indicators	4	3	2	1	Score (Maximum)
Form	Spelling Grammar Sentence structure	minimal errors and a high level of writing			many errors; poor writing skills	___ x 1 = ___(4)
Visual appeal and originality	Cover Artwork Graphics Different media	All 4 elements are creatively and visually appealing and original.	All 4 elements present	One element missing ; or lack of creativity	More than one element missing; boring presentation	___ x 1.5 = ___(6)
Organization	Completeness Timelines Table of Contents Pages numbered	All 4 elements demonstrate high level of organization	All 4 elements present	One element missing	More than one element missing	___ x 1 = ___(4)
Content and Knowledge of key concepts	Key concepts Evidence of understanding Application	Evidence of ability to apply knowledge to new situations	Evidence of high level of understanding of key concepts	Evidence of basic level of understanding of key concepts	Little evidence of key concepts included in portfolio	___ x 2 = ___(8)

TOTAL (22)

Rubric 4**Student Reflection Rating Scale**

Scale: 4-Always 3-Usually 2-Occasionally 1-Seldom

1. Clear and careful link evident between reflection and artefact	4----3----2----1
2. In-depth, insightful, and critical analysis of learning	4----3----2----1
3. Detailed examples, references, and connections provided	4----3----2----1
4. Honest, realistic self-assessment	4----3----2----1
5. Evidence of progress toward achievement of goals	4----3----2----1
6. Future goals clearly outlined	4----3----2----1
7. Vivid impressions create a sense of connection with audience	4----3----2----1

TOTAL (28)

RUBRIC 5
Presentation (10%)

choice of topic	Poor (1) The topic is not interesting nor relevant. Perhaps too simple	Fair (2) The topic is fairly interesting and relevant, not too simple	Good (3) The topic is important and interesting but might be too complex	Excellent (4) The topic is very important and very interesting and relevant. Not too simple or too complex
Quality of slides	Poor (1) The presentation is not attractive and cluttered or too sparsely laid out.	Fair (2) The presentation is fairly attractive and laid out. There is lots of room for improvement	Good (3) The presentation is attractive and well laid out.	Excellent (4) The presentation is very attractive and well laid out
Communication skills	Poor (1) The student reads from the slides, does not have eye contact or connection with the audience	Fair (2)	Good (3)	Excellent (4) The student engages actively with the audience. Presentation skills show confidence.
Understanding of material presented	Poor (1) The understanding is weak	Fair (2) Fair understanding	Good (3) Student shows a good level of understanding	Excellent (4) Excellent understanding
Explanation of stats concepts	Poor (1) The student does not explain thoroughly to non statistical audience and uses terms that are too complex	Fair (2)	Good (3)	Excellent (4) The student explains well to non statistical audience and uses terms that are at the right level of complexity
Handling of questions	Poor (1) The student does not invite questions or does not answer the questions posed.	Fair (2)	Good (3)	Excellent (4) The student actively invites questions and answers them very well.
Overall impression	Poor (1) the presentation leaves a lot to be desired.	Fair (3) the presentation could be improved	Good (5) the presentation is good.	Excellent (6) The presentation cannot be faulted and is of excellent quality.

PORTFOLIO TOTAL (Assignment 1+ Assignment 2+ Rubric 3+ Rubric 4 + Rubric 5+ Assignment 3) = 100

*** Guide to write up of reflections**

This is self-evaluation and experiential learning, where we want you to write a bit about what you learnt and how it has influenced specific tasks.

For reflections on lectures, look at the lecture objectives and critically assess whether objectives were met. Do you feel that you are at the point you should be after the lecture? How could the lecture have been improved? How will you use the information from the lecture to add to your proposal, or not? Do you think you will encounter any difficulties in using the information from the lecture to improve your study design? Can this new knowledge be applied more widely elsewhere in your life?

Reflections on each lecture can be as long as you want, but should not be less than 300 words. You can write it up in any style you feel comfortable with: point form, essay form, paragraphs with subheadings etc.

See the link below for more detail about reflection and a suggested structure for a reflective report.

<http://www.uefap.com/writing/genre/reflect.htm>

References

1. Friedman Ben David, M, Davis M.H, Harden R.M, Howie P.W, Ker J, Pippard MJ. AMEE Medical Education Guide no.24: Portfolios as a method of student assessment. *Medical Teacher*. 2001. 23 (6):535-551